



Making cities greener

Task A: The importance of urban greening

- 1) Imagine you were the mayor of a medium-sized city, and were given the task of adding two different **urban** greening strategies of your choice to the city.
- a) Which two strategies do you think would be the most cost effective to implement? Explain why.

Any answer may be valid with the correct justification.

Examples include:

- *Cycle lanes - some roads may be made one-way only, allowing the other half of the road to be turned into cycle lanes. No new cycle lanes need to be built.*
- *Trees in streets - tree saplings can be planted along different streets.*

- b) Which two strategies do you think would be most effective at improving the well-being of the population of the city? Explain why.

Examples include:

- *Green parks - people in the city will have easier-to-access green spaces to enjoy walks, a relaxing space for their lunch, etc.*
- *Improve public transport options - people may worry less about travelling to work on time on public transport. Options for travel are increased, or became more reliable.*

- c) Which two strategies do you think would be most effective at improving the sustainability of the city? Explain why.

Examples include:

- *Improve public transport options - fewer people using cars on their own means a reduction in carbon emissions. This is especially true if the public transport uses electric vehicles.*
- *Insulated buildings - the cost and energy consumption when heating a house during the winter can be very high.*

Note: You can use the same strategy for more than one question.

- 2) Several old cities in the UK have tried to implement **urban** greening strategies, but have struggled to do so. Give one possible reason why.

*Old cities that have grown larger over time may not have grown with **urban** greening in mind. This means these cities may not have the space for urban green parks, or the roads may not be wide enough to accommodate cycle lanes, routes for large buses, trains or trams, etc.*



Task B: The maths behind urban greening

- 1) The **urban** city of Oakfield has been given £ 10 000 000 of funding to start an **urban** greening initiative.

transport carbon emissions	461.3 kt	planting trees along streets (requires average of 3.5 hours of daily sunshine)	£17 000 per km ²
population	102 504		
total area of city	98 km ²	cycle lanes (suggested only if transport carbon emissions per person are over 0.004 kt)	£4 100 000
area available for renewal	23 km ²	urban park renewal (requires 25% of city area to be available)	£35 000 per km ²
area of bus depot.	7500 m ²	electric, zero-emission buses (area of 200 m ² required per bus)	£320 000 per bus
yearly total sunshine hours	1356 hours		

Using the tables of information about Oakfield and the cost of each **urban** greening strategy, recommend a set of effective **urban** greening strategies (and how much of the city it will apply to) that Oakfield can afford to implement.

The quantity of transport carbon emission per person is: 0.0045 kt (2 s.f.)

The percentage area of the city available for renewal is 23% (2 s.f.)

The average daily amount of sunshine is 3.7 hours (2 s.f.)

planting trees along streets (requires average of 3.5 hours of daily sunshine)	£17 000 per km ²	<i>At 3.7 hours per day average, this strategy is available.</i>
cycle lanes (suggested only if transport carbon emissions per person are over 0.004 kt)	£4 100 000	<i>At 0.0045 kt per person, this strategy is available.</i>
urban park renewal (requires 25% of city area to be available)	£35 000 per km ²	<i>At only 23% area available, this strategy is not available.</i>
electric, zero-emission buses (area of 200 m ² required per bus)	£320 000 per bus	<i>A maximum of 37 buses can fit in the depot.</i>

Oakfield cannot afford all three strategies, as their total cost is £17 606 000. However, it is possible to have the cycle lanes and all the street trees, plus 13 electric buses (at £4 160 000). Or the cycle path, some street trees, and a greater number of electric buses.

- 2) Discuss possible limitations of your proposed **urban** greening initiative in question 1.

A large assumption is that the entire area of Oakfield is “street” that can have trees planted on them. In reality, the area of Oakfield that is “street” might be a lot less than 98 km², since a lot of the land will either be buildings or roads.

Another assumption is that all the roads are wide enough, and off-road locations are safe and unobstructed enough, to have cycle lanes built on them.

